



# CSEE4119: Computer Networks

## Final Preview

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Slides credited to J.F Kurose and K.W. Ross and their textbook "Computer Networking: A Top-Down Approach"

# Recap

## ■ Basics

- Protocol
- Circuit switching vs packet switching
- Performance metrics: transmission rate/time, packet delay, end-to-end throughput, packet loss rate, bandwidth latency product, propagation delays, etc
- Layers: modular functionality, headers

# Recap

- Application layer
  - Client-server, peer-to-peer
  - Sockets
  - Web: persistent, non-persistent HTTP connection; web cache
  - DNS
  - Video streaming
  - P2P: BitTorrent, file distribution time analysis, DHT

# Recap

- Transport layer
  - UDP: connection-less, minimalist
  - TCP: Reliable data transfer
    - Mechanisms to deal with bit errors, packet delay, packet losses, out-of-order arrivals, duplicated packets
    - GBN vs selective repeat

# Recap

- Transport layer
  - TCP: Congestion control
    - End-host based vs network-assisted congestion control
    - TCP main mechanisms: slow start, AIMD
    - Packet-loss based vs packet-delay based
    - Mainstream: CUBIC, BBR

# Recap

- Network layer: data plane
  - Longest-prefix matching
  - IP address: prefix/subnet, host address, DHCP
  - Network address translation (NAT)
  - IPv4 vs IPv6

# Recap

- Network layer: control plane
  - Two types of routing algorithms: link state routing, distance vector routing
  - Their comparison
  - BGP, difference from distance vector routing
  - SDN concept

# Recap

- Data link layer for wired links
  - Error detection and correction code
  - Medium access control
    - Goal, types
    - ALOHA, efficiency analysis
    - CSMA/CD



# Recap

- Wireless
  - Basics of wireless physical layer: free-space pathloss model, characteristics of wireless channel
  - Wireless MAC protocols
    - Hidden terminals, exposed terminals
    - CSMA/CA

## Example Question

#1: To TCP, a lost segment is interpreted as a sign of congestion: if an ACK is not received by the sender, the sender will reduce its rate of transmission.

In which scenario does a lost segment **not** imply congestion?

**A:** An ACK would not be sent if the segment has bit errors and does not pass the checksum. In this case, it does not imply congestion.

## Example Question

#2: The dynamic host configuration protocol (DHCP) is the network management protocol used by devices to request IP addresses from the local DHCP server. Suppose your computer receives a reply from the DHCP server with an IP address 128.16.51.2 and a subnet mask 255.255.240.0.

What is the network prefix of your newly assigned IP address? What is the host address? How many hosts can be in the subnet?

**A:** The network prefix is 128.16.48.0 or 128.16.48.0/20, and the host address is 0.0.3.2. Twelve bits are used to represent the host address, so  $2^{12}$  hosts can be in the subnet.

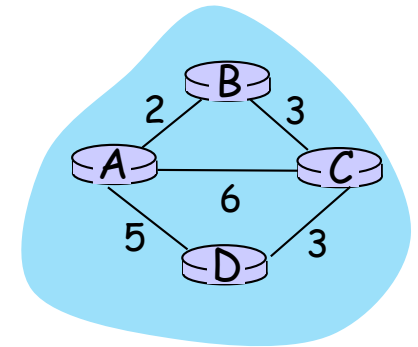
## Example Question

#3: Columbia has a direct network connection to CERN in Geneva, for collaboration with the Large Hadron Collider team. Columbia also has a peering relationship with NYU. Can NYU students go to CERN via Columbia? If not, what aspect of BGP can be used to make sure that everyone on the Columbia network can get to CERN but nobody else can get to CERN via Columbia?

**A:** No, NYU students cannot get to CERN via Columbia. BGP path export policy will allow Columbia not to share its direction path to CERN to NYU.

## Example Question

#4: Four nodes form a network topology as below. When they run the distance vector routing algorithm, what is the initial distance vector table at A and B? What is the converged distance vector at A and B respectively, after exchanging distance vectors with neighbors?



**A:**

A's initial DV table

|   | B        | C        | D        |
|---|----------|----------|----------|
| B | 2        | $\infty$ | $\infty$ |
| C | $\infty$ | 6        | $\infty$ |
| D | $\infty$ | $\infty$ | 5        |

B's initial DV table

|   | A        | C        |
|---|----------|----------|
| A | 2        | $\infty$ |
| C | $\infty$ | 3        |

A's final DV

| Dst | min |
|-----|-----|
| B   | 2   |
| C   | 5   |
| D   | 5   |

B's final DV

| Dst | min |
|-----|-----|
| A   | 2   |
| C   | 3   |
| D   | 6   |

## Example Question

#5: Four nodes are located as shown in the right figure. The circles illustrate nodes' carrier sensing ranges. When node A is transmitting data to B, who are the potential hidden terminals and exposed terminals?

**A:** Hidden terminal: C, E, no exposed terminal.

